Below is the revised syllabus based on student comments. The original syllabus can be found at Original Syllabus. The annotated syllabus with student comments can be found here.

Lecture	M/T/W/Th 8:30-11:00, see Canvas > Zoom for URL
Instructor	Professor Pisan **Email**: pisan@uw.edu
Office Hours	Make an appointment using Calendar > Find Appointment Regular slots Tuesday 4-5pm and Friday 1-2pm. Email me with multiple suggested dates and times if you cannot make these times.

Course Description

Principal ideas and developments in artificial intelligence, such as problem solving, knowledge representation, search, reasoning under uncertainty, learning, and natural language processing.

We will use python (specifically python 2.7) for the programming assignments.

Course Learning Goals

ABET Outcomes: (high level goals that the course contributes to)

1. an ability to apply knowledge of mathematics, science, and engineering;

2. an ability to design a system, component, or process to meet desired needs within

3. realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability;

4. an ability to use the techniques, skills, and modern engineering tools;

5. knowledge of contemporary issues;

6. a recognition of the need for, and an ability to engage in lifelong learning;

7. an ability to identify, formulate, and solve engineering problems; and

8. an ability to design and conduct experiments, as well as to analyze and interpret data

At the end of this course, students will be able to:

1. Describe what artificial intelligence (AI) means and how machines can process information intelligently;

2. Understand the role of machine learning within the larger field of artificial intelligence;

3. Identify the different fields that comprise AI as well as techniques and heuristics for each area;

4. Create a formal representation of a complex problem; and

5. Write programs to solve complex problems using AI methods in areas such as search, reasoning, natural language processing, games, and robotics.

Textbooks

(Recommended) Russell, Stuart J., and Norvig, Peter. *Artificial Intelligence : a Modern Approach*. 3rd edition, Pearson, 2015. http://aima.cs.berkeley.edu/ and https://www.amazon.com/Artificial-Intelligence-Approach-Stuart-Russell/dp/9332543518

For additional books and references see https://github.com/pisanorg/w/wiki books You are also encouraged to contribute new links to the wiki.

Grading

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. 2.0 roughly corresponds to 75%. The above percentage grades are the minimum required to get the decimal mark indicated. If you get 80%, you are guaranteed to get at least 2.5. In general, grades are not curved, so your success is not determined by how well or badly other students in the class are doing.

- Exercises: 15% - can miss one without any penalty

    - Exercises are small assessments from the lower levels of Bloom's taxonomy. These exercises test whether you remember and understand the concepts discussed and whether you can apply them in novel situations. We will have in-class exercises, syllabus annotation exercise, Canvas quizzes, short group exercises and other activities as part of this group of assessment tasks.

- Projects: 65%

  - Projects are substantial assessment items from the higher levels of the Bloom's taxonomy. Projects require you to analyze and evaluate new information as well as create original work. Projects take significant time and effort. Most, but not all, of our projects in this course will be implementing algorithms that have been discussed in class and coming up with new solutions.

- Final: 20%

  - The final is an open-book, open-internet, 2-hour exam with questions from the higher levels of the Bloom taxonomy.

Policies

Attendance: Attend all classes. If you are going to miss a class, I'd appreciate a courtesy email with an explanation, but it is not required. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications.

While there is no penalty for missing a class, missing a class might also mean you will miss one or more graded Exercises. Your lowest mark in "Exercises" will be automatically dropped (see under Canvas > Assignments to see which of the assignments are under the "Exercises" heading). If you know that you will be missing several classes, you should reconsider dropping this course.

If you miss a class, you need to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc. Emailing me to ask if "there was anything important', expecting me to repeat the lecture or summarize it for you is considered rude. You are welcome to ask clarifying questions once you have made a good faith effort to understand the material using the available resources.

Staying focused during online courses is challenging, but there are several things you can do to keep focused:

- When possible, find a quiet room where you can work undisturbed for the duration of the lecture

- Close all browser windows or documents open that are not relevant for the current class activity

- Close your email account, log out of social media accounts

- Keep your camera on. If you do not have a camera, you can login to Zoom simultaneously through your desktop computer and your phone.

- Make sure you have a Zoom profile picture, so that if your camera is not working properly, we can still see (and remember) what you look like. You can use a professional picture, such as one from your LinkedIn profile, or an informal picture.

- Keep an eye on the Zoom chat and participate as appropriate

- Participate fully in breakout room exercises. If somebody in your breakout room is not participating in the exercise, let me know via email. I will monitor the situation. If somebody is failing to contribute over multiple sessions, I will contact them to address the issue.

- For some people, having something to eat or drink can also help focus. Be mindful of your choices, so you are not eating lots of unhealthy snacks and compromising your health at the expense of trying to remain focused.

An important aspect of the course is creating a supportive learning environment, finding and connecting with peers that you can reach out when you are stuck. Breakout rooms give you opportunities to work with your classmates. Being able to see each other, even when it is a tiny picture on Zoom, strengthens these social connections.

While there might be a good reason to take a screenshot of the lecture notes being shared, there is no good reason to take a screenshot of Zoom participants, just like you don't usually take pictures of the class when you attend an in-person class. If you can’t resist the awkward screenshot, do NOT post it on the internet without getting permission from all the people in the screenshot. Regardless of your intentions, once posted any image can be misinterpreted, modified, go viral, and can cause real harm.

I understand that you may need to abruptly leave the Zoom session to attend to any number of unexpected events from a barking dog to a child needing attention. Very few people have a private room where they won't be interrupted for their Zoom sessions. Do the best you can. Attend to any emergencies, take care of your family. If you leave the Zoom session for an extended period of time, review the recordings and send me an email afterwards.

Assignment Submission: Late assignments are not accepted. Your lowest mark in "Exercises" will be automatically dropped (see under Canvas > Assignments to see which of the assignments are under the "Exercises" heading). Follow the assignment submission instructions as each assignment will have different requirements.

I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Effort: This course is offered as a Term-B Summer Quarter course. Expect to spend 20-30 hours per week outside class. If an assignment is worth 10%, expect to spend 10-20 hours on the assignment. If you are spending too much time or too little time, let me know we'll adjust the course content. Learning happens best when your are challenged and get to stretch your limits.

Exam Procedures: Final exam will be open-book, open-internet during the last scheduled lecture for Term-B Summer Quarter. I will provide sample questions to help you prepare for the exam.

Academic Integrity: Do the right thing, see http://guides.lib.uw.edu/bothell/ai for details on what is "right" if in doubt. Talking about code is OK, looking at each others code is not OK. Looking at references to understand how a function gets used is OK; searching the internet for assignment solutions is not OK.

Communication: Join the #382 channel on discord https://discord.gg/5mEm92e Discord is an extension of the classroom, so use a meaningful nickname and act professionally. If your question can be answered by a classmate, post it to discord rather than using email or waiting for office hours. If your question needs my attention, tag me on discord. Office hours are for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" if you need to communicate directly with me on issues that cannot be discussed on discord. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, make an appointment and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Access and Accommodations: Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course. See http://www.uwb.edu/studentaffairs/drs if you need to establish accommodations.

(The following paragraph has been written by lawyers and I have been asked to include it as is in the course syllabus. It does not match the language of the rest of the syllabus. What is important is that I will make reasonable accommodations for religious and other reasons. Let me know your circumstances as soon as possible, and we can work on a solution.)

Religious Accommodations: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Religious Accommodations Policy (https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/). Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form (https://registrar.washington.edu/students/religious-accommodations-request/).

Common Course Policies for the School of STEM: See https://www.uwb.edu/stem/about/stem-policies for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, support for Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity. Student Support Services, Surviving Sexual and Relationship Violence, Wondering How to Address Faculty? etc. (P.S. I prefer to be addressed as 'Professor Pisan')

Course Material: Lecture notes and other material will be posted to Weekly Notes after the lecture.

Conversation Café

To help you connect with others in the course, we've created a completely optional, synchronous "Conversation Café." The café is an informal time/space that will be open to you every Wednesday from 3-4pm (Pacific). There is no agenda; it is not instructor-driven; it is not office hours. It's simply a place where you can chat, catch up, and/or share your experiences with other people in the course. To enter the café, click on the Conversation Café link in the Course Navigation Menu to the left.

Topics

See Weekly Notes for details and lecture notes.

Week-1

- Thursday, July 23, 2020: Introduction

Week-2

- Monday, July 27, 2020: Uninformed Search
- Tuesday, July 28, 2020: Informed Search
- Wednesday, July 29, 2020: Constraint Satisfaction Problems I
- Thursday, July 30, 2020: Constraint Satisfaction Problems II

Week-3

- Monday, August 3, 2020: Adversarial Search I
- Tuesday, August 4, 2020: Adversarial Search II
- Wednesday, August 5, 2020: Expectimax Search and Utilities
- Thursday, August 6, 2020: Markov Decision Problems I

Week-4

- Monday, August 10, 2020: Markov Decision Problems II
- Tuesday, August 11, 2020: Reinforcement Learning I
- Wednesday, August 12, 2020: Reinforcement Learning II
- Thursday, August 13, 2020: Review

Week-5

- Monday, August 17, 2020: Markov Models
- Tuesday, August 18, 2020: Hidden Markov Models
- Wednesday, August 19, 2020: Review

- Thursday, August 20, 2020: Online Final